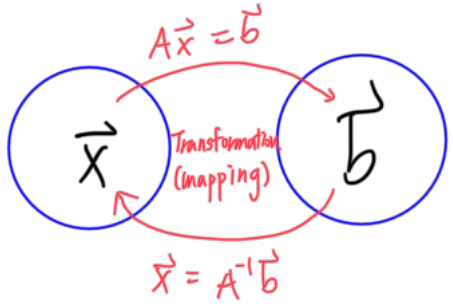
Linear Algebra
(113-2, 2025 Spring)

LINEAR TRANSFORMATION 1.8, 1.9, 2.3

Linear Transformations

- Linear Transformations 1.8
- Matrix Transformations 1.9
- Characterization of Invertible Matrices 2.3



Transformations

A **transformation** (or **function** or **mapping**) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.

T maps $\mathbb{R}^n$ to $\mathbb{R}^m$

$$T: \mathbb{R}^n \to \mathbb{R}^m$$

$$\boxed{\mathbf{x} \mapsto T(\mathbf{x})} \in \mathbb{R}^m, \text{for each } \mathbf{x} \in \mathbb{R}^n$$

map $\vec{x}$ in $\mathbb{R}^n$ to $T(\vec{x})$ in $\mathbb{R}^m$

The set $\mathbb{R}^n$ is called **domain** of T, and $\mathbb{R}^m$ is called the **codomain** of T.

$\vec{x}$ in $\mathbb{R}^2$

if $T = \begin{bmatrix} 3 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \cdots$

$T(\vec{x}) = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ in $\mathbb{R}^3$ space

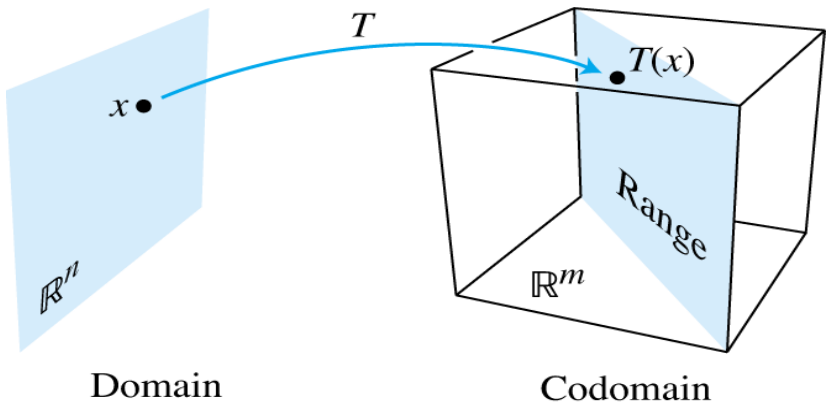
Range of T is a line (in $\mathbb{R}^3$)

- For $\mathbf{x}$ in $\mathbb{R}^n$, the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is called the **image** of $\mathbf{x}$ (under the action of T).

映射

- The set of all images $T(\mathbf{x})$ is called the **range** of T.

A subset of $\mathbb{R}^m$



Domain, codomain, and range of $T: \mathbb{R}^n \to \mathbb{R}^m$.

Matrix Transformations

$T: \mathbb{R}^n \to \mathbb{R}^m$

$A = m \times n$

For each $\mathbf{x}$ in $\mathbb{R}^n$, $T(\mathbf{x})$ is computed as $A\mathbf{x}$, where A is an $m \times n$ matrix.

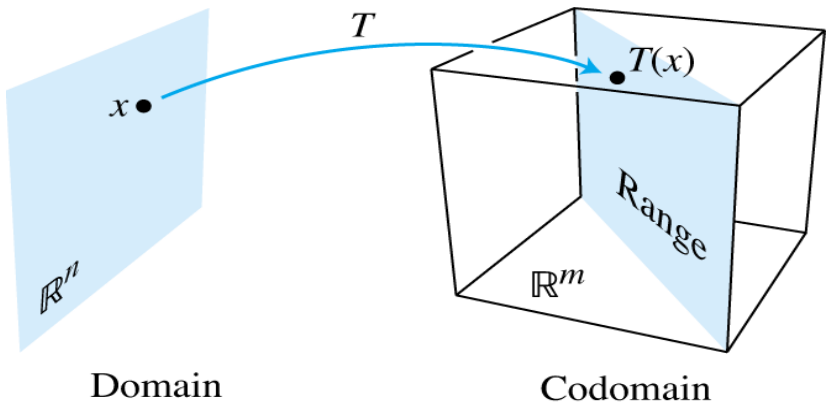
For simplicity, we denote such a *matrix transformation* by $\mathbf{x} \mapsto A\mathbf{x}$. $m \times 1 \in \mathbb{R}^m$

$$A = m \underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}}_{n}$$

Observe that the domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when each column of A has m entries.

- The range of T is the set of all linear combinations of the columns of A, because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$. $A: m \times n$ $\vec{x}: n \times 1$ as $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

$A\vec{x} = x_1\vec{a}_1 + x_2\vec{a}_2 + x_3\vec{a}_3 \cdots x_n\vec{a}_n$



Domain, codomain, and range of $T: \mathbb{R}^n \to \mathbb{R}^m$.

Matrix Transformations (Cont'd)

Example 1:

$T: \mathbb{R}^2 \to \mathbb{R}^3$ by $T(\vec{x}) = A\vec{x}$

Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$,

$\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$

d. $\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{array}\right]$ Inconsistent

and define a transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

$$= x_1 \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 5 \\ 7 \end{bmatrix}$$

a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T. $\begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$

b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$. What x can give you b $T(\vec{x}) = \vec{b} = A\vec{x} = \vec{b}$

c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$? If $T(x_1) = \vec{b}$, $T(x_2) = \vec{b}$, $\vec{x}_1 \neq \vec{x}_2$ $A\vec{x} = \vec{b}$ has many solutions

d. Determine if $\mathbf{c}$ is in the range of the transformation T.

b. $\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{array}\right]$ $x_1 = \frac{3}{2}$? $x_2 = -\frac{1}{2}$

$\vec{c} \in \text{Range}(T) = \{A\vec{x}\}$ all linear combinations

Shear Transformation

Example 3: ☆ p. 102, 103 special transformation

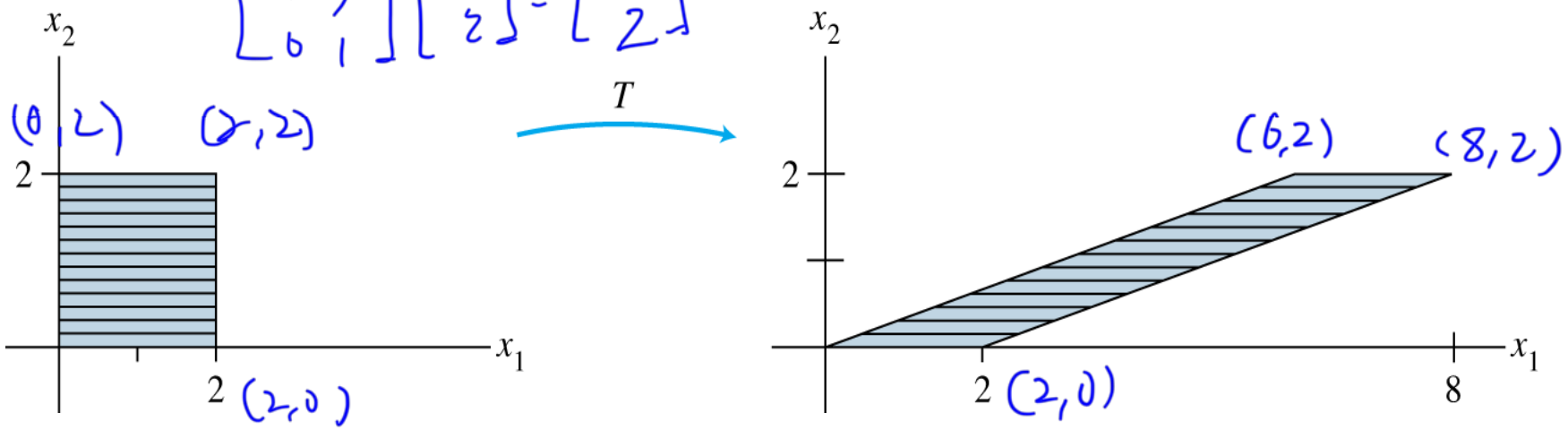
Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. The transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$ is called a shear transformation

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

a. The image of point $\mathbf{u} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

b. The image of point $\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.



Linear Transformations

Definition:

A transformation (or mapping) T is **linear** if:

1. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all **u**, **v** in the domain of T
2. $T(c\mathbf{u}) = cT(\mathbf{u})$for all scalars c and all **u** in the domain of T.

- Linear transformations *preserve the operations of vector addition and scalar multiplication*.

- $T: \mathbb{R}^n \to \mathbb{R}^m$ is linear.
  1) $T(\mathbf{0}) = \mathbf{0}$.
  2) $T(-\boldsymbol{v}) = -T(\boldsymbol{v})$.
  3) $T(\boldsymbol{u}-\boldsymbol{v}) = T(\boldsymbol{u}) - T(\boldsymbol{v})$.
  4) $T(\alpha\mathbf{u}+\beta\boldsymbol{v}) = \alpha T(\boldsymbol{u}) + \beta T(\boldsymbol{v})$.

- Generalization:
  $T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p)$
  $= c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + \cdots + c_pT(\mathbf{u}_p)$
  ➔ *superposition principle* in engineering and physics.
  - Think of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ as signals that go into a system and $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_p)$ as the responses of that system to the signals.

Linear Transformations (Cont'd)

Example:

1. $T: \mathbb{R}^2 \to \mathbb{R}^2$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_2 + 2 \end{bmatrix}$. Is T linear?

2. $T: \mathbb{R}^2 \to \mathbb{R}^2$ by $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} |a| \\ b \end{bmatrix}$. Is T linear?

3. $T: \mathbb{R}^3 \to \mathbb{R}^2$ by $T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a - 2b \\ 3b + c \end{bmatrix}$. Is T linear?

Linear Transformations (Cont'd)

Given a scalar r, define $T: \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{x}) = r\mathbf{x}$.

- T is called a **contraction** when $0 \leq r \leq 1$ and a **dilation** when $r > 1$.

Example:

$T: \mathbb{R}^3 \to \mathbb{R}^2$ is linear.

$T\left(\begin{bmatrix}1\\2\\1\end{bmatrix}\right) = \begin{bmatrix}1\\0\end{bmatrix}, T\left(\begin{bmatrix}2\\9\\0\end{bmatrix}\right) = \begin{bmatrix}-1\\1\end{bmatrix},$

$T\left(\begin{bmatrix}3\\3\\4\end{bmatrix}\right) = \begin{bmatrix}0\\1\end{bmatrix}$. What is $T\left(\begin{bmatrix}7\\13\\7\end{bmatrix}\right)$?

Linear Transformations (Cont'd)

Example:

$T: \mathbb{R}^2 \to \mathbb{R}^2$ is linear that rotate $\theta°$.

What is $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$?

Linear Transformations

- Linear Transformations 1.8
- Matrix Transformations 1.9
- Characterization of Invertible Matrices 2.3

The Matrix of a Linear Transformation

THEOREM 10

Let T: $\mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

$$A = [T(e_1) \ \dots T(e_n)]_{m\times n}$$

- The matrix A is called the **standard matrix for the linear transformation** T.

pf.

$$\begin{aligned} &T(x) \\ &= T(x_1e_1 + \cdots + x_ne_n) \\ &= x_1T(e_1) + \cdots + x_nT(e_n) \\ &= [T(e_1) \quad \dots \quad T(e_n)]\begin{bmatrix} x_1 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} \\ &= Ax \end{aligned}$$

The Matrix of a Linear Transformation (Cont'd)

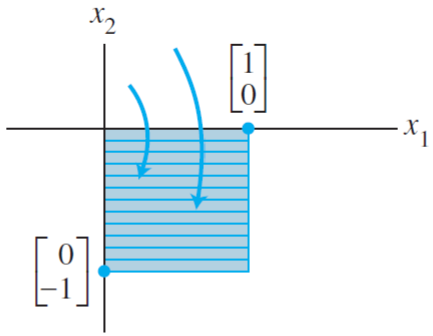
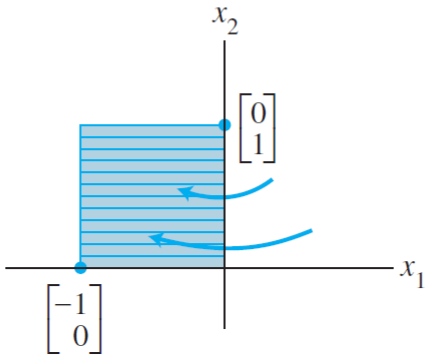
- We know now that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be viewed as a matrix transformation, and vice versa.
- The term *linear transformation* focuses on a property of a mapping, while *matrix transformation* describes how such a mapping is implemented, as the example illustrates.

Example 2:

Find the standard matrix *A* for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for **x** in $\mathbb{R}^2$.

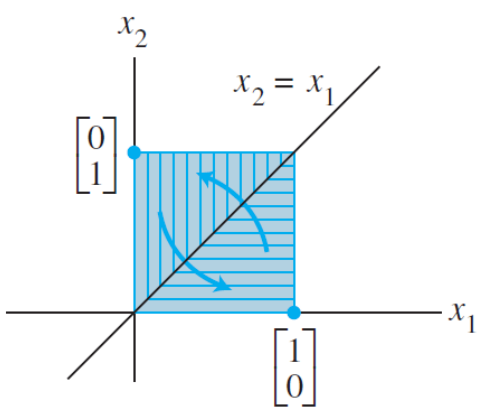
The Matrix of a Linear Transformation (Cont'd)

TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1-axis		$\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$
Reflection through the x_2-axis		$\begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}$

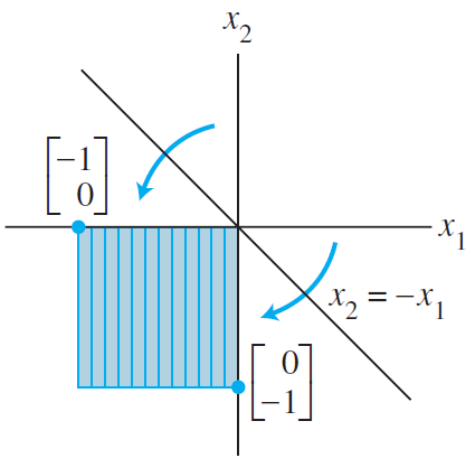
The Matrix of a Linear Transformation (Cont'd)

Reflection through the line $x_2 = x_1$



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

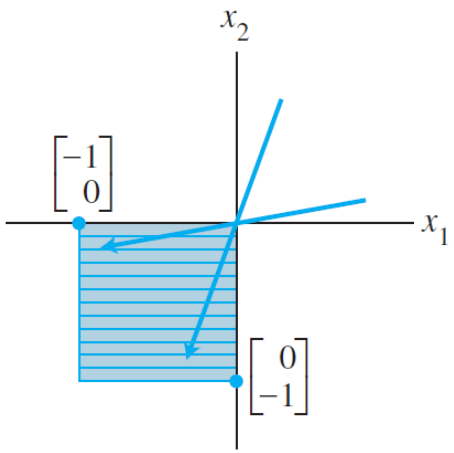
Reflection through the line $x_2 = -x_1$



$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

The Matrix of a Linear Transformation (Cont'd)

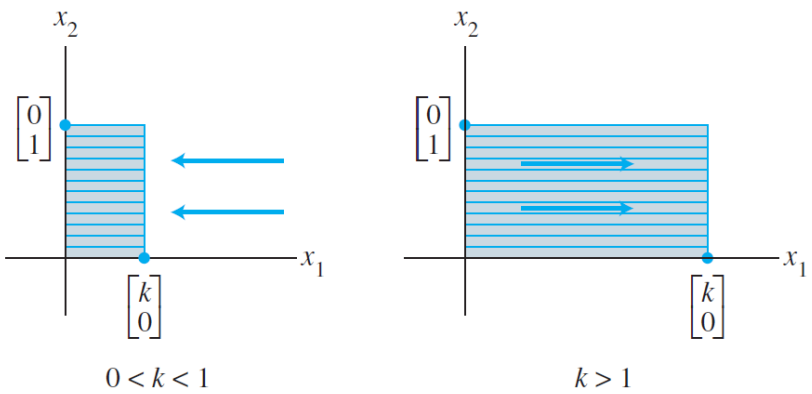
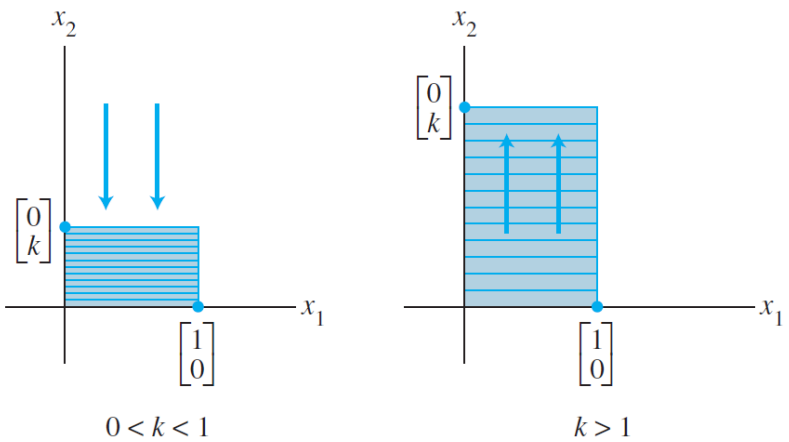
Reflection through the origin



$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

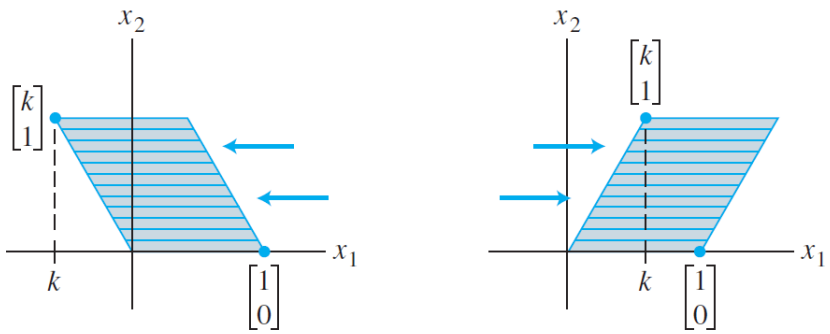
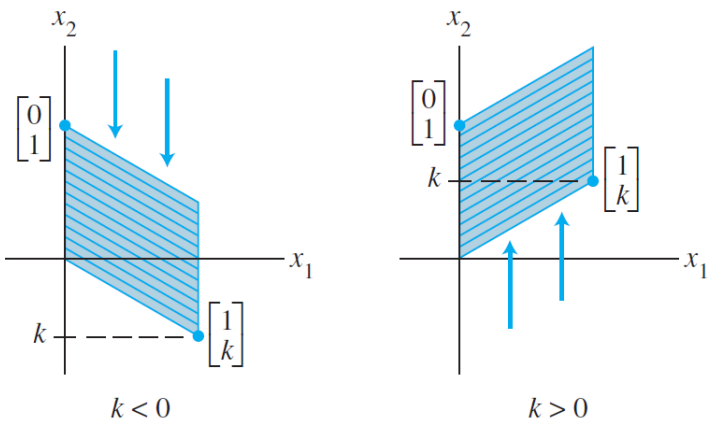
The Matrix of a Linear Transformation (Cont'd)

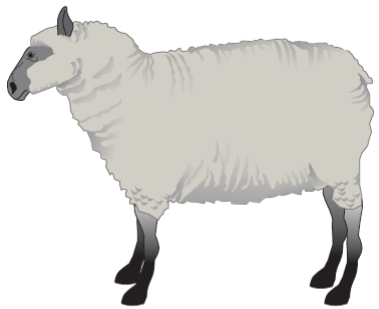
TABLE 2 Contractions and Expansions

Transformation	Image of the Unit Square	Standard Matrix
Horizontal contraction and expansion		$\begin{bmatrix}k & 0\\0 & 1\end{bmatrix}$
Vertical contraction and expansion		$\begin{bmatrix}1 & 0\\0 & k\end{bmatrix}$

The Matrix of a Linear Transformation (Cont'd)

TABLE 3 Shears

Transformation	Image of the Unit Square	Standard Matrix
Horizontal shear		$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
Vertical shear		$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

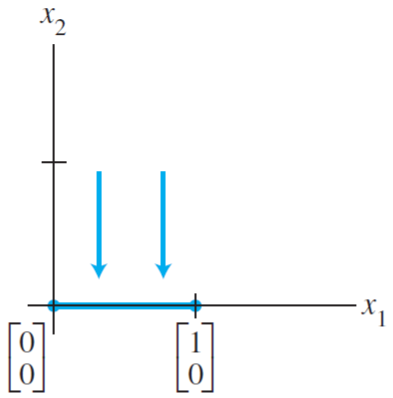
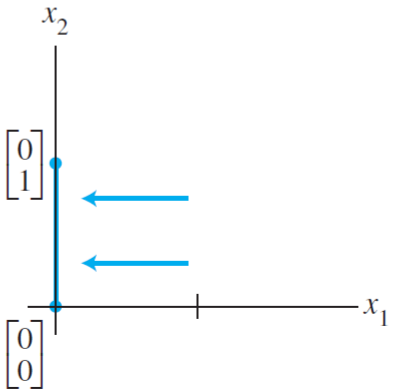


sheep

sheared sheep

The Matrix of a Linear Transformation (Cont'd)

TABLE 4 Projections

Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_1-axis		$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$
Projection onto the x_2-axis		$\begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}$

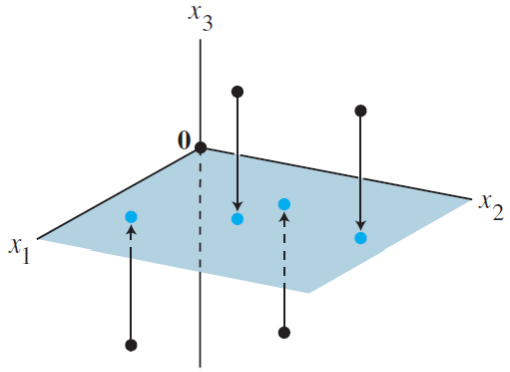
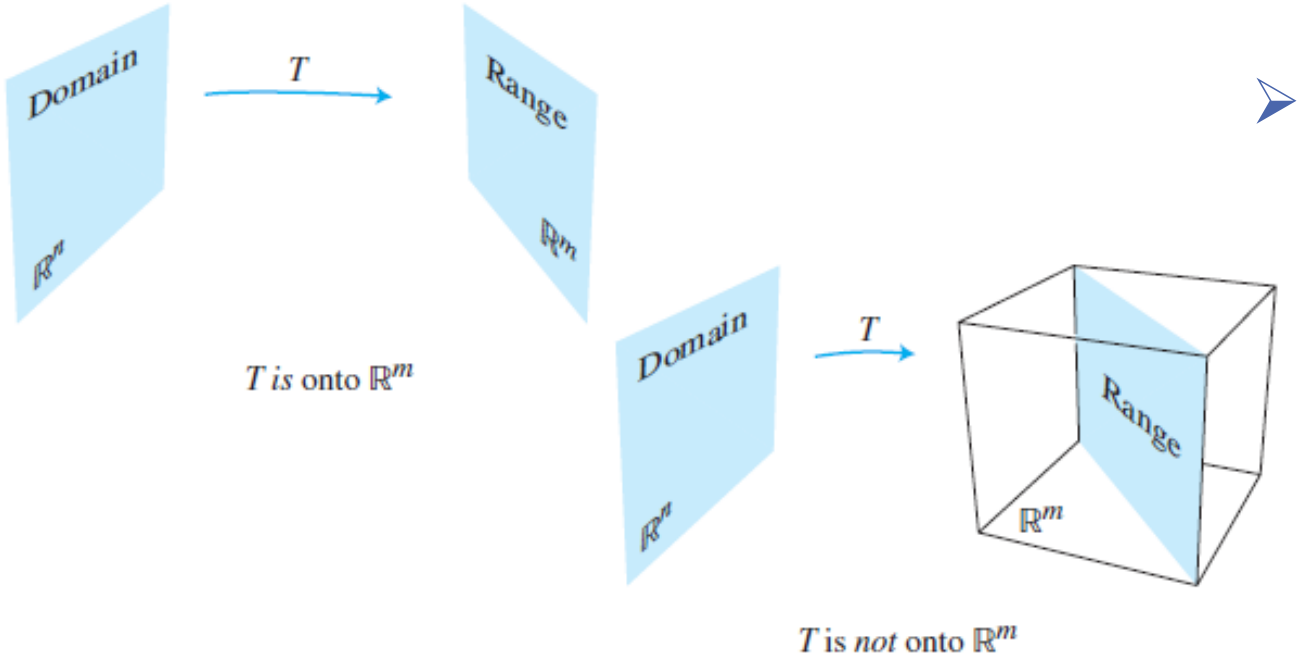


FIGURE 3
A projection transformation.

Existence and Uniqueness Questions

Definition:

A mapping T: $\mathbb{R}^n \to \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each **b** in $\mathbb{R}^m$ is the image of *at least one* **x** in $\mathbb{R}^n$.



- Equivalently, T is onto $\mathbb{R}^m$ when the range of *T* is all of the codomain $\mathbb{R}^m$.
  That is, *T* maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if, for each **b** in the codomain $\mathbb{R}^m$, there exists at least one solution of $T(\mathbf{x}) = \mathbf{b}$.

- "Does T map $\mathbb{R}^n$ onto $\mathbb{R}^m$?" is an existence question. The mapping *T* is not onto when there is some **b** in $\mathbb{R}^m$ for which the equation $T(\mathbf{x}) = \mathbf{b}$ has no solution.

Existence and Uniqueness Questions (Cont'd)

Definition:

A mapping T: $\mathbb{R}^n \to \mathbb{R}^m$ is said to be **one-to-one** if each **b** in $\mathbb{R}^m$ is the image of *at most one* **x** in $\mathbb{R}^n$.

Example 4: Let T be the linear transformation whose standard matrix is

$$A=\begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

1. Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$?
2. Is *T* a one-to-one mapping?

Existence and Uniqueness Questions (Cont'd)

THEOREM 11

Let T: $\mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

pf.

Since *T* is linear, T(0) =0. If *T* is one-to-one, then the equation $T(\mathbf{x}) = \mathbf{0}$ has at most one solution and hence only the trivial solution.

If *T* is not one-to-one, then there is a b that is the image of at least two different vectors in $\mathbb{R}^n$-- say, **u** and **v**. That is $T(\mathbf{u}) = \mathbf{b}$ and $T(\mathbf{v}) = \mathbf{b}$. But then, since *T* is linear,

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

The vector **u – v** is not zero, since **u ≠ v**. Hence the equation $T(\mathbf{x}) = \mathbf{0}$ has more than one solution. So, either the two conditions in the theorem are both true or they are both false.

Existence and Uniqueness Questions (Cont'd)

THEOREM 12

Let T: $\mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation and let A be the standard matrix for T. Then:

1. *T* maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of *A* span $\mathbb{R}^m$;
2. *T* is one-to-one if and only if the columns of *A* are linearly independent.

pf.

1. The columns of A span $\mathbb{R}^m$
   $\Leftrightarrow \forall \mathbf{b} \in \mathbb{R}^m$ s.t. $A\mathbf{x} = \mathbf{b}$ is consistent.
   $\Leftrightarrow A\mathbf{x} = T(\mathbf{x}) = \mathbf{b}$ has solutions.
   $\Leftrightarrow T$ is onto $\mathbb{R}^m$.

2. T is one-to-one.
   $\Leftrightarrow T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.
   $\Leftrightarrow A\mathbf{x} \neq \mathbf{0}, \forall \mathbf{x} \neq \mathbf{0}$
   $\Leftrightarrow$ The columns of A are linearly independent.

Existence and Uniqueness Questions (Cont'd)

Example:

$$T: \mathbb{R}^2 \to \mathbb{R}^3 \; by \; T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} a - 2b \\ 3a + b \\ a + b \end{bmatrix}$$

(1) Is T one-to-one?

(2) Is T onto $\mathbb{R}^3$?

Linear Transformations

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The Invertible Matrix Theorem

THEOREM 8

Let $A: n \times n$ be a linear transformation. Then Then the following statements are equivalent. That is, for a given *A*, the statements are either all true or all false.

a. *A* is an invertible matrix.
b. *A* is row equivalent to the $n \times n$ identity matrix.
c. *A* has *n* pivot positions.
d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
e. The columns of *A* form a linearly independent set.
f. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
g. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each b in $\mathbb{R}^n$.
h. The columns of *A* span $\mathbb{R}^n$.
i. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
j. There is an $n \times n$ matrix *C* such that $CA = I$.
k. There is an $n \times n$ matrix *D* such that $AD = I$.
l. A^T is an invertible matrix.

The Invertible Matrix Theorem (Cont'd)

Proof:

- If statement (a) is true, then A^{-1} works for C in (j), so $(a) \Rightarrow (j)$.
- Next, $(j) \Rightarrow (d)$.
- Also, $(d) \Rightarrow (c)$.
- If A is square and has *n* pivot positions, then the pivots must lie on the main diagonal, in which case the reduced echelon form of A is I_n.
- Thus, $(c) \Rightarrow (b)$.
- Also, $(b) \Rightarrow (a)$.

$$\begin{tikzcd}
 & (a) \arrow[dr, Rightarrow] & \\
(b) \arrow[ur, Rightarrow] & & (j) \arrow[d, Rightarrow] \\
(c) \arrow[u, Rightarrow] & & (d) \arrow[ll, Rightarrow]
\end{tikzcd}$$

- $(a) \Rightarrow (k)$ because A^{-1} works for D.
- Also, $(k) \Rightarrow (g)$ and $(g) \Rightarrow (a)$.
- So (k) and (g) are linked to the circle.
- Further, (g), (h), and (i) are equivalent for any matrix.
- Thus, (h) and (i) are linked through (g) to the circle.
- Since (d) is linked to the circle, so are (e) and (f), because (d), (e), and (f) are all equivalent for *any* matrix A.
- Finally, $(a) \Rightarrow (l)$ and $(l) \Rightarrow (a)$.

The Invertible Matrix Theorem (Cont'd)

Theorem 8 could also be written as "The equation $A\mathbf{x} = \mathbf{b}$ has a *unique* solution for each **b** in $\mathbb{R}^n$."

- This statement implies (b) $A \sim I_n$ and implies A : invertible.

Let A and B be square matrices. If $AB = I$, then A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$.

The Invertible Matrix Theorem divides the set of all $n \times n$ matrices into two disjoint classes:

- the invertible (nonsingular) matrices
- the noninvertible (singular) matrices.
  - A: singular $\Rightarrow A \nsim I_n$
    $\Rightarrow$ The columns of A are linearly dependent.

[Note]: The Invertible Matrix Theorem applies only to square matrices.

The Invertible Matrix Theorem (Cont'd)

Example 1:

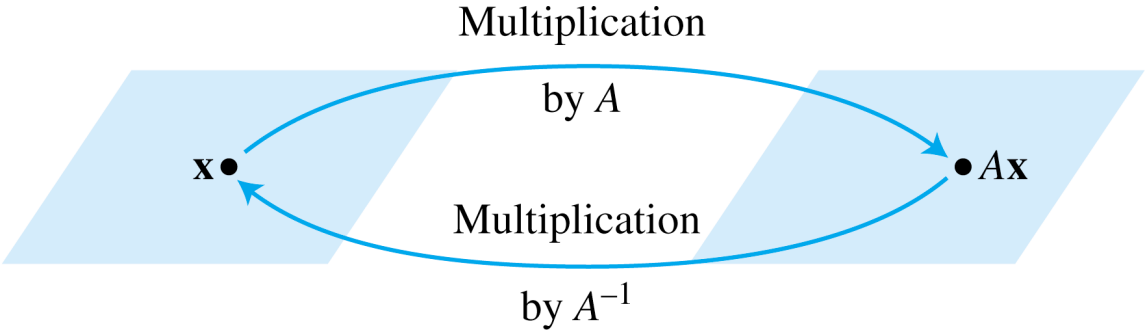
Use the Invertible Matrix Theorem to decide if A is invertible:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

Sol.

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

A has three pivot positions and hence is invertible, by the Invertible Matrix Theorem, statement (c).



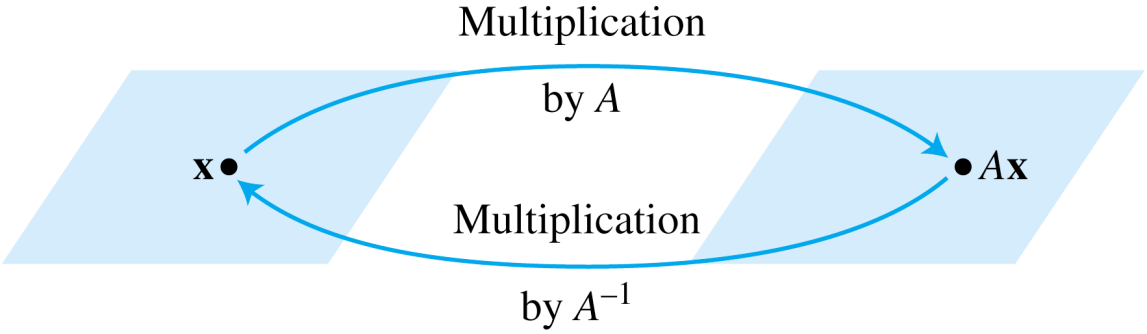
A^{-1} transforms $A\mathbf{x}$ back to $\mathbf{x}$.

When a matrix A is invertible, the equation $A^{-1}A\mathbf{x} = \mathbf{x}$ can be viewed as a statement about linear transformations.

The Invertible Matrix Theorem (Cont'd)

A linear transformation $T: \mathbb{R}^n \to \mathbb{R}^n$ is said to be **invertible** if there exists a function S: $\mathbb{R}^n \to \mathbb{R}^n$ such that

$$S\big(T(\mathbf{x})\big) = \mathbf{x},\ \forall \mathbf{x} \in \mathbb{R}^n \quad (1)$$
$$T\big(S(\mathbf{x})\big) = \mathbf{x},\ \forall \mathbf{x} \in \mathbb{R}^n \quad (2)$$



A^{-1} transforms $A\mathbf{x}$ back to $\mathbf{x}$.

THEOREM 9 (Invertible Linear Transformation)

Let $T: \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation and let <u>A be the standard matrix for T</u>. Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(\mathbf{x}) = A^{-1}(\mathbf{x})$ is the unique function satisfying equation (1) and (2).

Exercise 41

T is a linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^2$. Show that T is invertible and find a formula for T^{-1}.

$$T(x_1, x_2) = (-9x_1 + 7x_2, 4x_1 - 3x_2)$$

Sol.

The standard matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2)] = \begin{bmatrix} -9 & 7 \\ 4 & -3 \end{bmatrix}$$

And, $B = A^{-1} = \frac{1}{-1}\begin{bmatrix} -3 & -7 \\ -4 & -9 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 4 & 9 \end{bmatrix}$.

So that, $T^{-1}(\mathbf{y}) = B\mathbf{y}, \mathbf{y} \in \mathbb{R}^2$

Assignment 05

1.8 – 2,4,8,20

1.9 – 2,20

2.3 – 6,7,42

Due 05/06 Tue. In-class